

ROVER LAB, DOSSIER (WASP-76 b mission)

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Planet

Name: WASP-76 b

Planet type: Ultra-hot, inflated gas giant ('hot Jupiter'), no solid surface

Surface temperature: Dayside $\approx 2400\text{--}2500\text{ }^{\circ}\text{C}$ (measured); nightside $\approx 1500\text{ }^{\circ}\text{C}$ (measured). Tidally locked.

Surface pressure: No solid surface, pressure increases with depth in the atmosphere; a 1 bar reference level is used by convention only.

Gravity: $\approx 0.7\text{ g}$ (surprisingly low for its mass, the planet is so puffed up by stellar heating that its radius is huge)

Atmosphere: Hydrogen/helium dominated, with confirmed atomic iron, titanium oxide, sodium, magnesium, calcium and manganese vapour detected (ESPRESSO, 2020–2022)

Mission Briefing

WASP-76 b is a gas giant so close to its star that its dayside atmosphere reaches temperatures hot enough to vaporise iron. Astronomers have confirmed real iron vapour streaming from the scorching dayside toward the cooler nightside on winds of nearly 18,000 km/h, a phenomenon popularly described as 'iron rain'. Mission control has deployed an atmospheric probe, 'Skyhook-1', to skim the twilight boundary between day and night and collect any liquid or condensed droplets it can find, hoping to recover a sample rich enough in iron to be useful for radiation shielding back at the station. There is no solid ground here, whatever the probe brings back will have to be judged purely by what it contains.

Rover

Starting energy: 100

Site visit cost: 10

Experiment cost: 15

Overview Image

Site: Amber droplets collected at the twilight cloud deck

Description: A sample of condensed liquid skimmed from swirling, glowing clouds at the boundary between the scorching dayside and the darker nightside.

Identity: Food-colourant solution (orange/amber dye in water), no real iron content. DECOY GROUP.

Known Properties

Colour: Amber-orange, similar in hue to a concentrated salt solution

Odour: None

Solubility: Fully miscible with water

Experiment Results

Measure pH: Close to neutral, pH around 6.5–7

Measure density: About the same as water, 1.0 g/mL

Test electrical conductivity: Low

Add a magnet: No attraction

Add potassium thiocyanate solution: No colour change

Add sodium hydroxide solution: No precipitate forms; the colour only fades slightly

Heat a small sample gently: No visible change

Experiments

Measure pH

Measure density

Test electrical conductivity

Add a magnet

Add potassium thiocyanate solution

Add sodium hydroxide solution

Heat a small sample gently

AI Behaviour

Reveal remaining candidates after narrowing: No

Socratic questions: Yes

Difficulty: Standard (15-16 years old)

Teacher Notes

DECOY GROUP, assign a food-colourant solution (orange/amber dye, no real iron), not real FeCl_3 . WASP-76b is real and its 'iron rain' is a genuine, published finding, which is exactly why it makes a good decoy: students may assume 'iron is real here, so the sample must be real iron', but the planet has no solid surface at all, so a rover could never actually retrieve a ground sample from it. Use this in the final conference discussion: even a planet with dramatic, confirmed iron chemistry can be a poor (or literally impossible) source of a physical iron sample, if there's no solid surface to sample from.

Real-science notes: WASP-76 b is a genuine, well-studied ultra-hot Jupiter $\sim 620\text{--}640$ light-years away (distance estimates vary slightly by source), discovered in 2013 (SuperWASP), orbiting its star in just 1.8 days. Ehrenreich et al. 2020 (Nature, ESPRESSO spectrograph) detected real atomic iron absorption, strong on the dayside and weak/absent on the nightside, interpreted as iron vapour condensing on the night side. Important nuance for your own understanding: the popular 'iron rain' description is a simplification; follow-up work (Wardenier et al. 2021) shows a simple day-night temperature difference could reproduce the same signal without literal condensation/rain, and high-altitude mineral clouds may also be involved. The iron detection itself is solid; the 'rain' mechanism is genuinely still debated, a nice real example of how a popular-science headline can outrun the nuance of the actual paper.

Sources: Ehrenreich et al. 2020, Nature ([nature.com/articles/s41586-020-2107-1](https://www.nature.com/articles/s41586-020-2107-1)); Wardenier et al. 2021, ApJ (iopscience.iop.org/article/10.3847/1538-4357/ac423f).

Cross-subject link: Biology selective pressure for this planet, (hypothetical) life adapted to floating in gas-giant clouds with $\sim 900^\circ\text{C}$ day-night temperature swings and iron-vapour-laden winds, a thought experiment in the spirit of real 'aerial biosphere' hypotheses proposed for gas giants.